

Implementation of Hybrid DNS Sinkhole Architecture Based on Pi-hole and Machine Learning in Mitigation of Online Gambling Site Access

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Table of Content

01 Introduction & Problem Identification

02 Threat Landscape, Attack Chain, & Objective

03 System Architecture, Flowchart, & Random Forest Evaluation

04 QoS Analysis & CPU Load Usage Scalability

05 Conclusion & Future Works

Introduction

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THE FINANCIAL CRISIS

Total turnover of the online gambling ecosystem in Indonesia during 2025.

12.3 M

TRAPPED PLAYERS

Active players caught inside the ecosystem across all age groups.

36,400

TARGETING MINORS

Children under 11 years old already exposed to gambling platforms.

THE TECHNICAL FAILURE

Conventional Defenses Still Rely On Static Blacklists.

They are entirely reactive: a domain is only blocked after it has been reported, verified and published. Automated domain regeneration produces new addresses far faster than any list can be updated, so the protection gap never closes.

Problem Identification

01 Economic

Massive capital outflow to overseas servers and weaker domestic purchasing power, pushing victims toward illegal online loans.

02 Manufacturability

Enterprise-grade ML firewalls demand expensive hardware, which puts them out of reach for home and small office networks.

03 Security

Attackers exploit DGA and hijack authoritative government domains, which renders static blocking useless.

04 Law

Takedown through legal bureaucracy is far slower than the generation rate of new gambling domains.

05 Social & Ethics

Open access drives severe addiction, damages family stability and correlates with rising crime rates.

THE CORE GAP

Blocking Must Become Predictive, Not Administrative.

A domain has to be judged by what it looks like, at the moment it is queried.

Domain Generation Algorithm (DGA)

Definition

A computational technique that dynamically generates thousands of pseudo-random domain names to serve as Command and Control (C2) rendezvous points.

The Security Threat

These domains regenerate periodically and stay active only for a very short lifespan, which makes conventional static blacklisting ineffective against zero-day domains.

Key Characteristics

DGA domains carry no natural linguistic meaning and show a very high level of character randomness.

Typical pattern: `kq7xvbz2ml.com` ·
`v9dfhq3ptw.xyz` · `z4mnbqxr7s.top`

The Countermeasure

Structural randomness can be measured. Machine learning sits at the front line and detects the anomaly through lexical & n-gram feature extraction.

How Gambling Syndicates Hijack Website Credibility

01

SEO Poisoning

Attackers manipulate search engine ranking to push gambling links to the top of the results, deliberately targeting high-authority government (.go.id) and educational (.ac.id) portals.

02

Web Defacement

After infiltrating a legitimate site they alter its interface or silently create hidden subdomains. The URL bar still shows a trusted root domain while the page serves a gambling interface.

03

Malicious Redirect

Obfuscated scripts or HTTP 301/302 responses on the compromised host forcefully redirect the browser to the real offshore gambling server, bypassing static filters.

Net effect: the destination never appears on a blacklist, and the victim only ever sees a trusted institutional URL so credibility itself becomes the delivery vehicle.

Proposed Solution

01 Low cost, low resource

A filtering system that runs on affordable hardware instead of enterprise appliances, so home and small networks can deploy it.

02 Real-time detection

Gambling and DGA domains are classified by machine learning at the moment the DNS query is made, not after a report is filed.

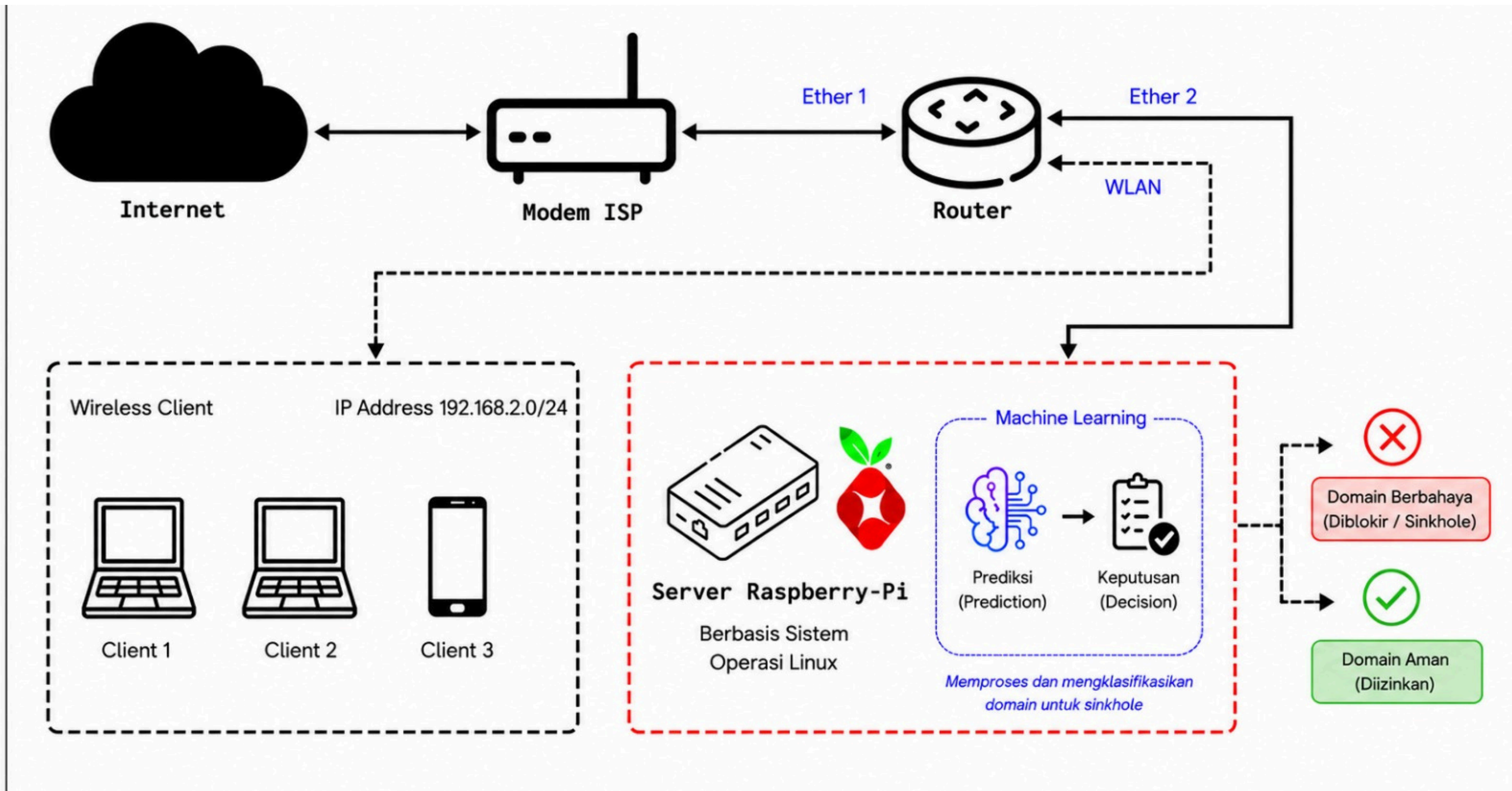
03 Self-updating database

Every domain the model flags is written back into the local database, so the blocklist grows automatically with the threat.

04 Accurate with minimal error

DGA domains are blocked aggressively while legitimate traffic stays usable, keeping false detections under the tolerance limit.

Topology of Architecture



1. Send out DNS Request

2. The Query Hits router

3. Forceful Interception

4. Analyzes & decision

5. Action Block / Forward

6. The system drop request

7. the sistem forward request

Flowchart of Architecture

01 Intercept

The router redirects every client DNS query to the proxy on port 53.

02 Cache lookup

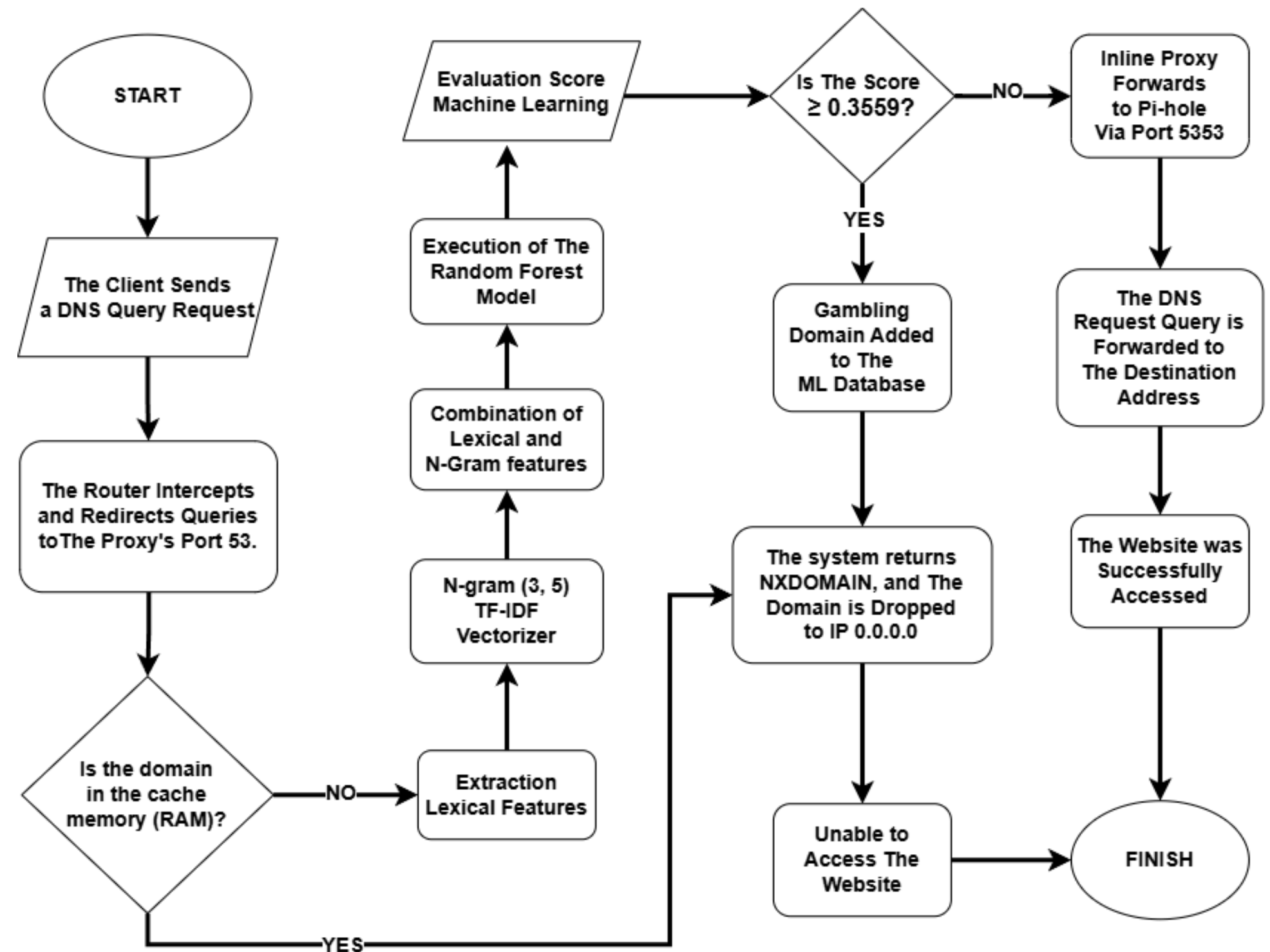
If the domain already exists in the SQLite database, the verdict is returned immediately.

03 Feature extraction

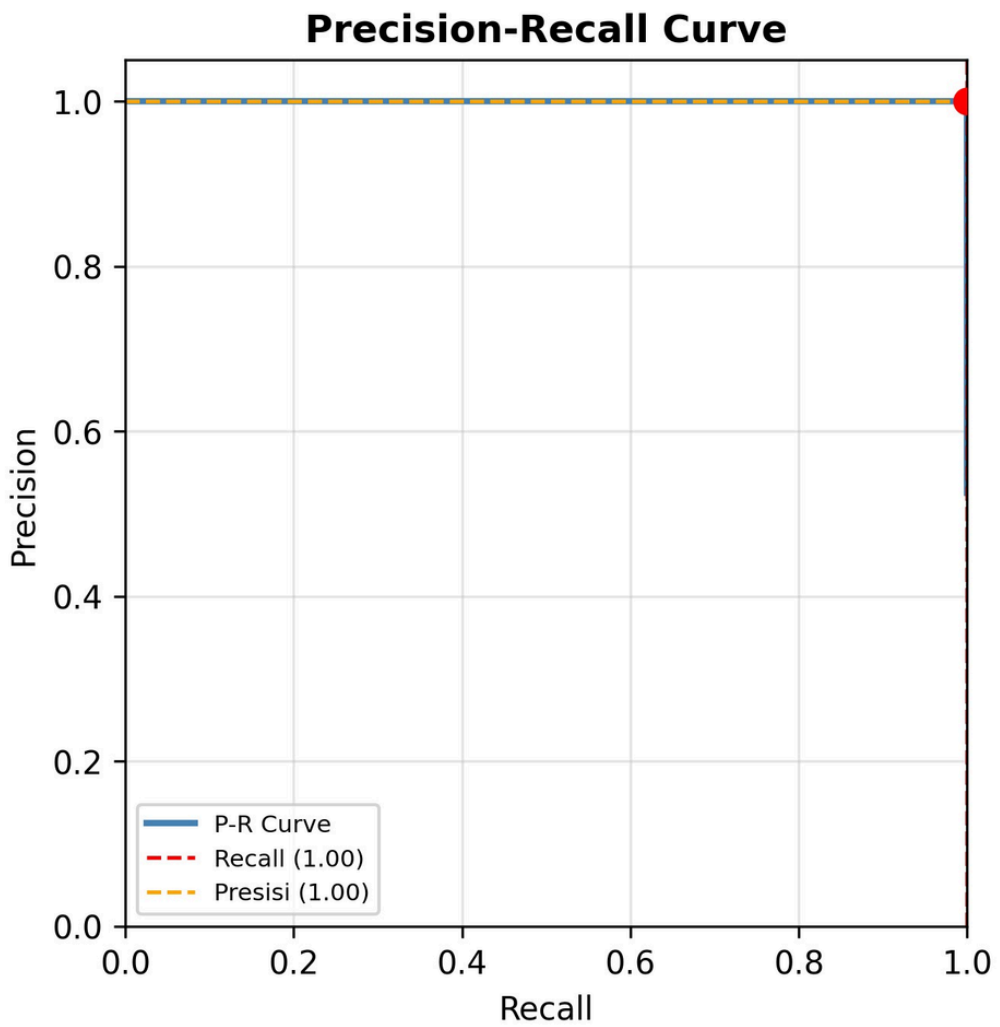
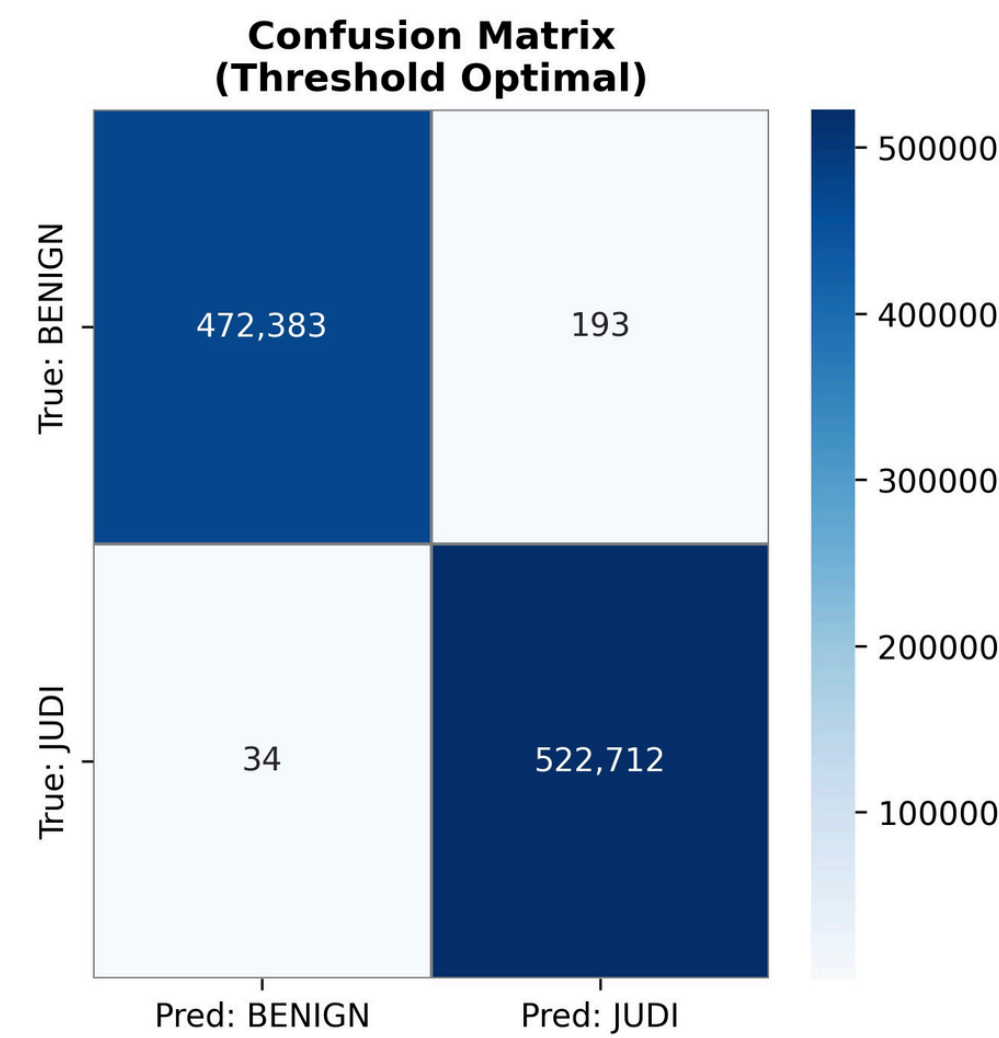
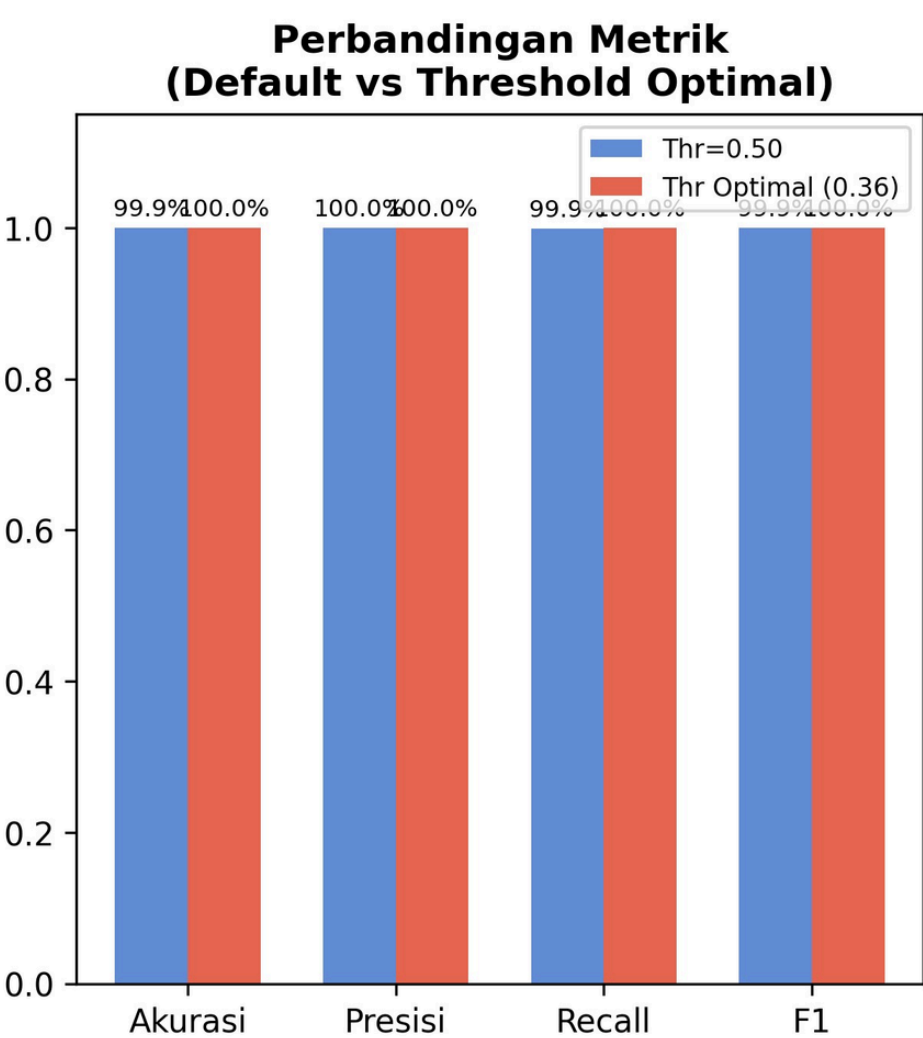
New domains go through lexical features and n-gram

04 Classify & act

Random Forest scores the domain; above 0.3559 it is sinkholed, otherwise Pi-hole handles it.

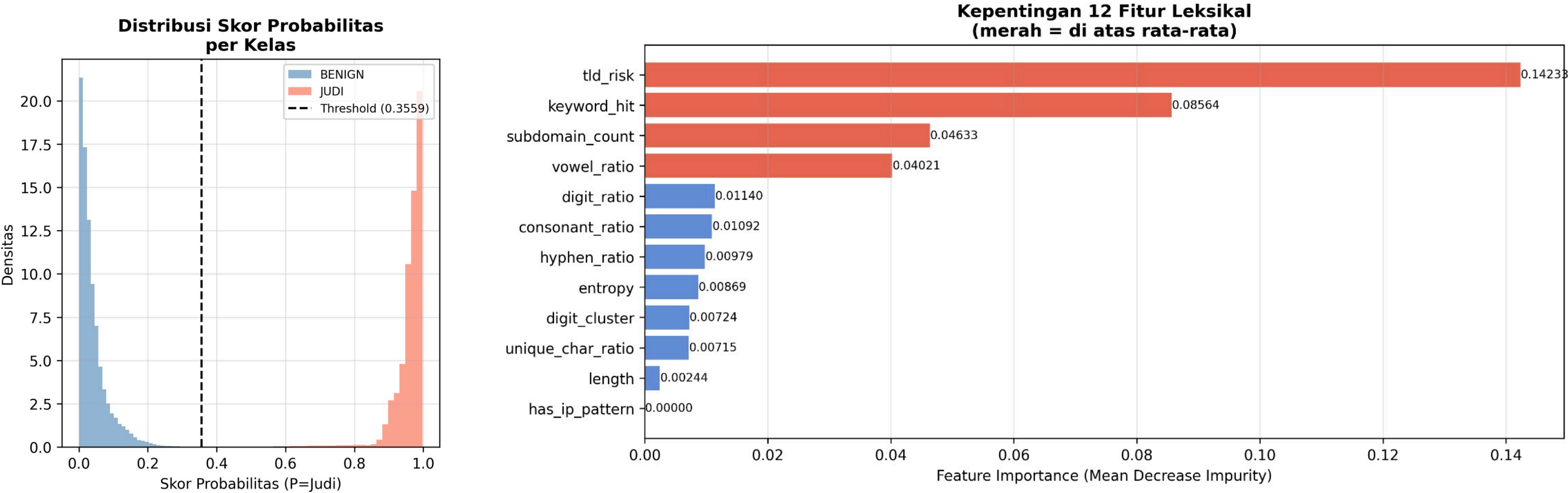


Random Forest Evaluation



Metric Comparison: Optimizing the threshold to 0.3559 maintained the stability of key metrics (Accuracy, Precision, Recall, F1) at an ideal equilibrium 99.9%, confirming the robustness of the model's vector space without the risk of overfitting. **Confusion Matrix:** The application of Cost-Sensitive Learning successfully suppressed the False Negative rate to an absolute minimum (34 out of hundreds of thousands of queries), aggressively prioritizing the isolation of cyber anomalies. **Precision-Recall Curve:** The asymptotic curve hugging the (1.0, 1.0) coordinate demonstrates the model's absolute superiority in handling data distribution imbalances.

Random Forest Evaluation



Probability Distribution: Radical bimodal polarization (BENIGN centered at 0.0 and JUDI at 1.0 demonstrates model convergence in the absence of classification uncertainty. **Feature Importance:** Gini impurity reduction demonstrates that deterministic linguistic parameters specifically `tld\_risk` and `keyword\_hit` act as primary discriminators in isolating the anomalous domain.

Detection Performance

Blocking Rate **98.00%**

$$TPR = \frac{294}{294 + 6} \times 100\% = 98,00 \%$$

Allowing Rate **95.33%**

$$TNR = \frac{286}{286 + 14} \times 100\% = 95,33 \%$$

USING 600 TOTAL TEST DOMAINS IN REAL-TIME

True Positive (TP)

294

Gambling domain,
correctly blocked

False Negative (FN)

6

Gambling domain,
missed

False Positive (FP)

14

Safe domain, wrongly
blocked

True Negative (TN)

286

Safe domain, correctly
allowed

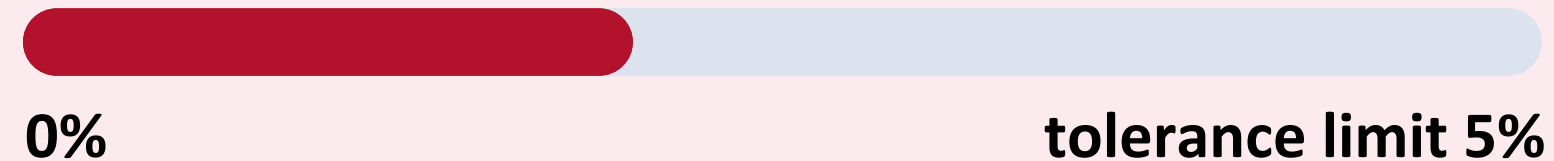
Test Set: 300 Malicious Domains + 300 Benign Domains.

Error Evaluation & Analysis

2.00% False Negative Rate (FNR)

$$FNR = \frac{6}{6 + 286} \times 100 \% = 2,00 \%$$

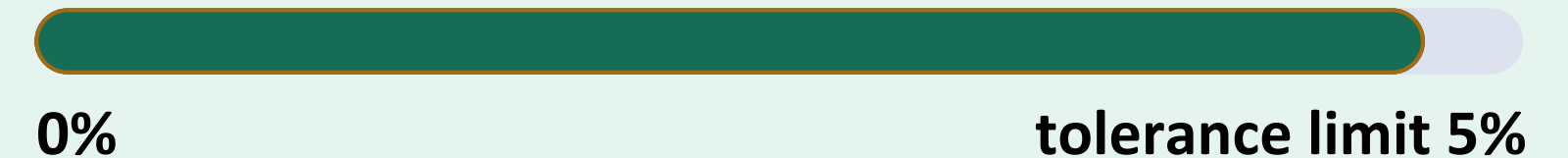
Suppressed to an extreme low of 2.00% (only 6 malicious domains bypassed the system).



4.67% False Positive Rate (FPR)

$$FPR = \frac{14}{14 + 286} \times 100 \% = 4,67 \%$$

Maintained at 4.67% (14 safe domains were incorrectly blocked).



Both error rates sit below the 5% tolerance target, so the aggressive filtering policy doesn't break everyday usability.

QoS Results & Analysis

Parameter	Baseline Daily Use	Baseline Stress Test	Sinkhole Daily Use	Sinkhole Stress Test
Minimum latency	1.76 ms	1.96 ms	3.82 ms	2.02 ms
Maximum latency	1294.87 ms	4241.41 ms	991.47 ms	4804.12 ms
Average latency	66.38 ms	348.75 ms	46.67 ms	416.39 ms
Throughput (QPS)	6.52	29.75	7.64	48,69
Queries lost	0%	0.83%	1.38%	23.22%

DAILY USE PERFORMANCE

- During the daily use scenario, the network maintains high efficiency with an average latency of 46.67 ms and a stable throughput capacity of 7.64 QPS.
- The system runs with high reliability, recording a minimal query loss rate of only 1.38% under normal operational loads.

STRESS TEST PERFORMANCE

- Under extreme traffic injection, the system yields an average latency of 416.39 ms while handling a maximum throughput peak of 48.69 QPS.
- The intense load triggers a query loss rate of 23.22%, which marks the ultimate operational and data processing boundary of the edge hardware.

CPU Load Usage Scalability

Devices	CPU Utilities Raspberry Pi	CPU Utilities Mikrotik
1	14.88 %	2,4 %
2	3729 %	3,3 %
3	43.11 %	3,9 %
4	46.52 %	4,9 %
5	44.88 %	5,9 %
6	41.05 %	6,4 %
7	46.85 %	8,2 %

CPU Utilities MikroTik

- **Load:** 2.4% – 8.2% (1 to 7 devices).
- **Efficiency:** Dst-NAT manipulates packet headers only; zero payload overhead.

CPU Utilities Raspberry Pi 4

- **Load:** Stabilized at 41% – 46%.
- **Optimization:** Fast-Path Cache instantly resolves repeated queries, bypassing redundant ML execution.
- **Stability:** Zero CPU saturation despite enforcing a strict zero blind-spot policy.

Final Conclusion

01 Optimal Threat Mitigation

98.00% for **Blocking Rate** and 95.33% for **Allowing Rate**, neutralising **Domain Generation Algorithm (DGA)** threats in real time.

02 Cost-sensitive Calibration

A threshold of 0.3559 suppresses the **False Negative Rate** to 2.00% while holding the **False Positive Rate** at a tolerable 4.67%.

03 Edge QoS Compliance

Maintains high efficiency with 46.67 ms **Latency** and only 1.38% Query Loss during **Daily Use**, while a 23.22% query loss at 48.69 **QPS** marks the ultimate operational boundary of the Edge Hardware, Raspberry Pi 4.

04 Computational Scalability

Fast-path caching stabilised CPU utilisation at 41–46% during peak inference load on the Raspberry Pi.

Suggestions for Future Works

01 Lexical Recalibration

Update the keyword corpus (domain_keyword.py) periodically to counter advanced lexical obfuscation and dilution tactics.

02 Proactive Administration

Enforce manual absolute whitelisting through the control panel so critical false positives can be rectified immediately.

03 Hardware Scaling

Upgrade proxy CPU specifications for enterprise deployment to remove the bottleneck during parallel n-gram processing.

04 Protocol Expansion

Add filtering modules for encrypted resolution **DNS over HTTPS (DoH)** and **DNS over TLS (DoT)**.

THANK YOU

Any Questions?